

암 환자의
비타민 D 보충제 섭취와
사망률에 대한
체계적 문헌 고찰 및 메타분석



식품생명공학과

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문헌 검색 전략 / 문헌 선택 과정

데이터 추출 / 통계 분석

결과

선정된 문헌 개요

암 진단 이후 비타민 D 보충과 Survival

고찰

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연구주제 설정 배경



서론

연구주제 설정 배경



선행 연구

비타민D
보충제

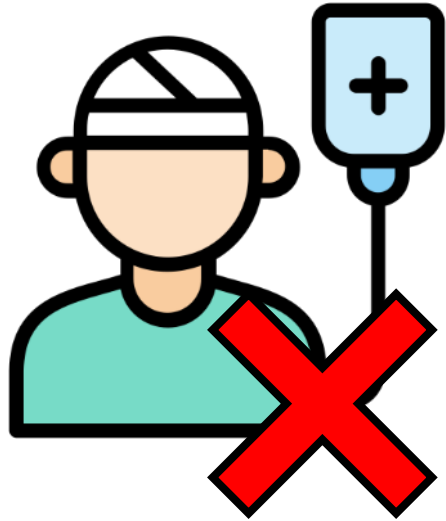
&

암 예방, 치료
메타분석

르사·크세·모·니·프

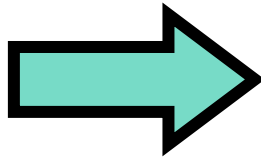
서론

연구주제 설정 배경



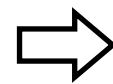
선행 연구

~~암 환자 대상~~ 전체 인구 대상
무작위 대조군 실험 연구 또는 코호트 연구에서
비타민 D 보충제 섭취와 사망률 등 결과 지표의
관계 추정 메타분석

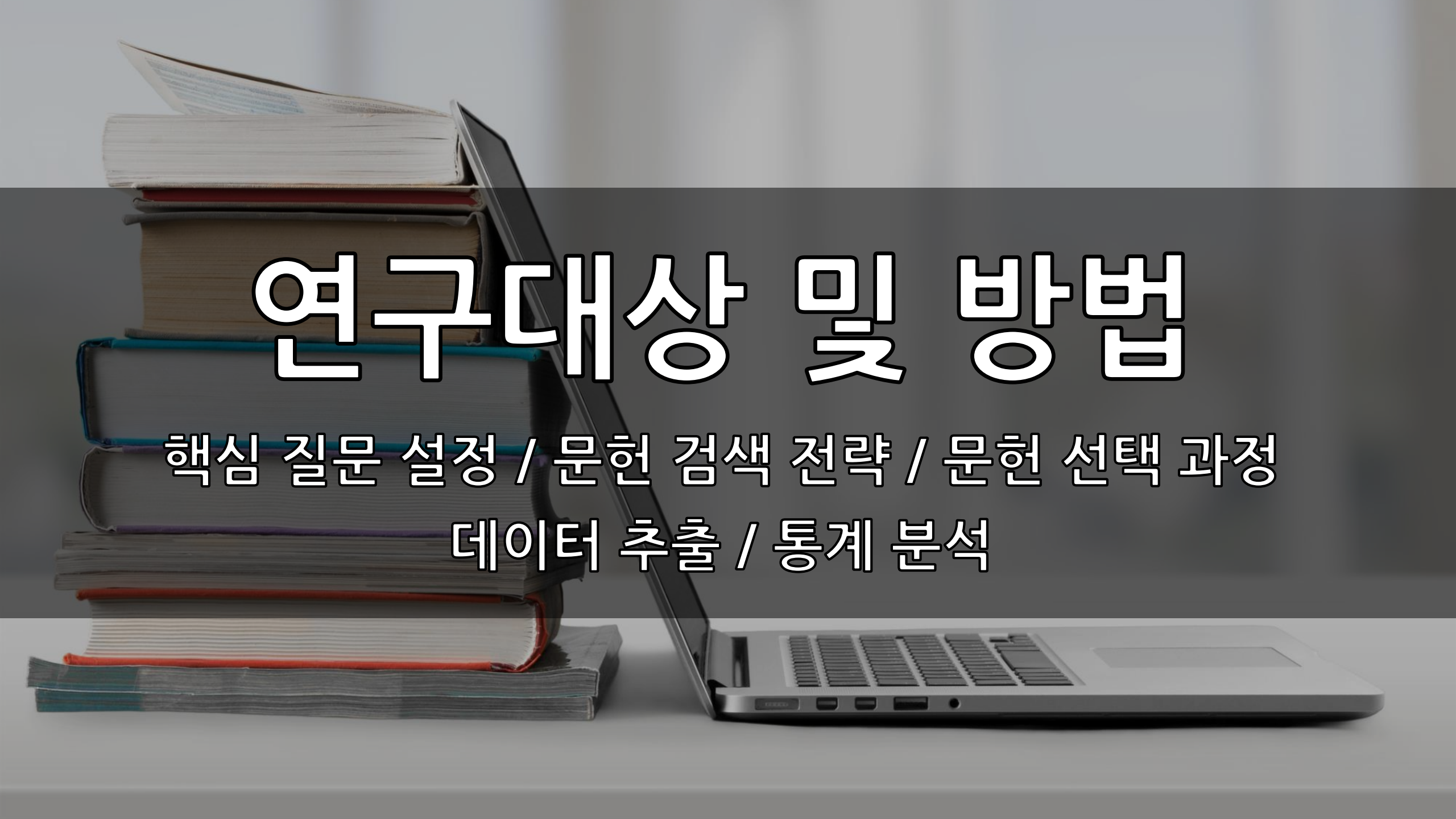


본 연구

암 환자 대상
무작위 대조군 실험 연구 또는 코호트 연구에서
비타민 D 보충제 섭취와 사망률 등 결과 지표의
관계 추정 메타분석



정보 제공 to 암환자:
비타민 D 보충제의 암 예방제로서 효능 유무

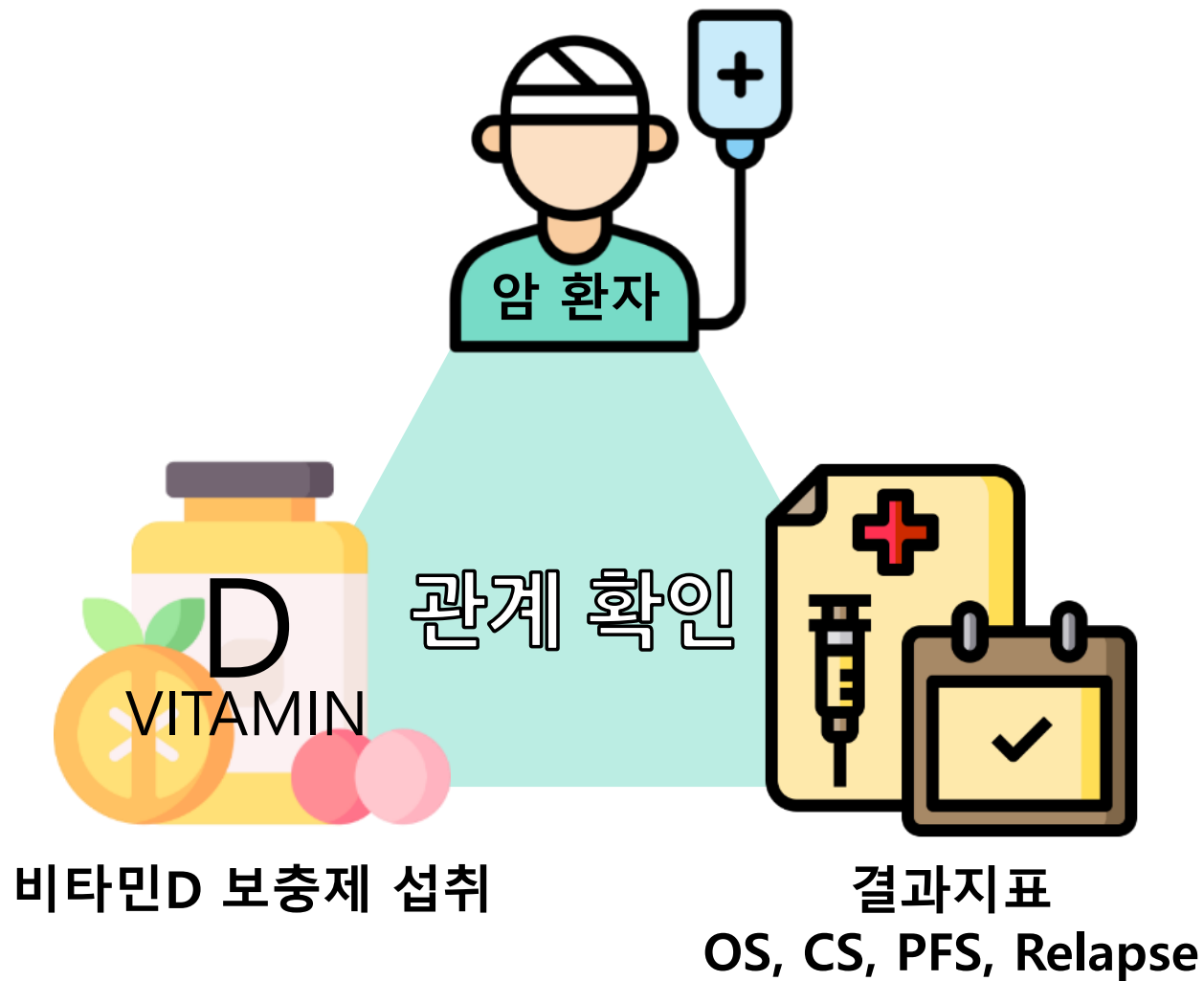
A stack of several books with various colored spines (white, red, blue, purple, orange) is on the left. A silver laptop is open to the right, with its screen partially visible. The background is a soft, out-of-focus grey.

연구대상 및 방법

핵심 질문 설정 / 문헌 검색 전략 / 문헌 선택 과정
데이터 추출 / 통계 분석

연구대상 및 방법

핵심 질문 설정



결과지표 정의

OS (Overall Survival):

환자가 치료를 받기 시작한 순간부터 사망한 순간까지의 기간

CS (Cancer Survival):

치료를 받기 시작한 순간부터 암으로 사망한 순간까지의 기간

PFS (Progression-free survival):

치료를 받기 시작한 순간부터 암의 재발 혹은 암으로 사망한 순간까지의 기간

Relapse:

치료를 받기 시작한 순간부터 암이 재발된 순간까지의 기간

연구대상 및 방법

문헌 검색 전략

문헌 검색



데이터 베이스
Pubmed, Embase

기준 기간

1983년 ~ 2022년 3월 14일

사용 검색어

Pubmed	("Vit D"[Mesh] OR Vit D[tw] OR cholecalciferol*[tw] OR ergocalciferol*[tw] OR "calcium"[Mesh] OR calcium[tw]) AND ("dietary supplements"[Mesh] OR supplement[tw] OR supplements[tw] OR supplementation[tw]) AND ("neoplasms"[Mesh] OR neoplasm[tw] OR neoplasms[tw] OR neoplasia[tw] OR cancer[tw] OR cancers[tw] OR tumor[tw] OR tumors[tw] OR tumour[tw] OR tumours[tw] OR carcinoma[tw] OR carcinomas[tw]) AND ("mortality"[Mesh] OR mortality[tw] OR death[tw] OR survival[tw]) NOT (Case Reports[ptyp] OR Comment[ptyp] OR Letter[ptyp] OR Editorial[ptyp]) NOT ("animals"[Mesh] NOT "humans"[Mesh])
Embase	('Vit d'/exp OR 'Vit D':ab,ti OR cholecalciferol:ab,ti OR ergocalciferol:ab,ti OR 'calcium'/exp OR calcium:ab,ti) AND ('dietary supplements'/exp OR supplement:ab,ti OR supplements:ab,ti OR supplementation:ab,ti) AND ('neoplasms'/exp OR neoplasm:ab,ti OR neoplasms:ab,ti OR neoplasia:ab,ti OR cancer:ab,ti OR cancers:ab,ti OR tumor:ab,ti OR tumors:ab,ti OR tumour:ab,ti OR tumours:ab,ti OR carcinoma:ab,ti OR carcinomas:ab,ti) AND ('mortality'/exp OR mortality:ab,ti OR death:ab,ti OR survival:ab,ti) NOT ('case report'/de OR 'comment'/de OR 'letter'/de OR 'editorial'/de) NOT ([animals]/lim NOT [humans]/lim)

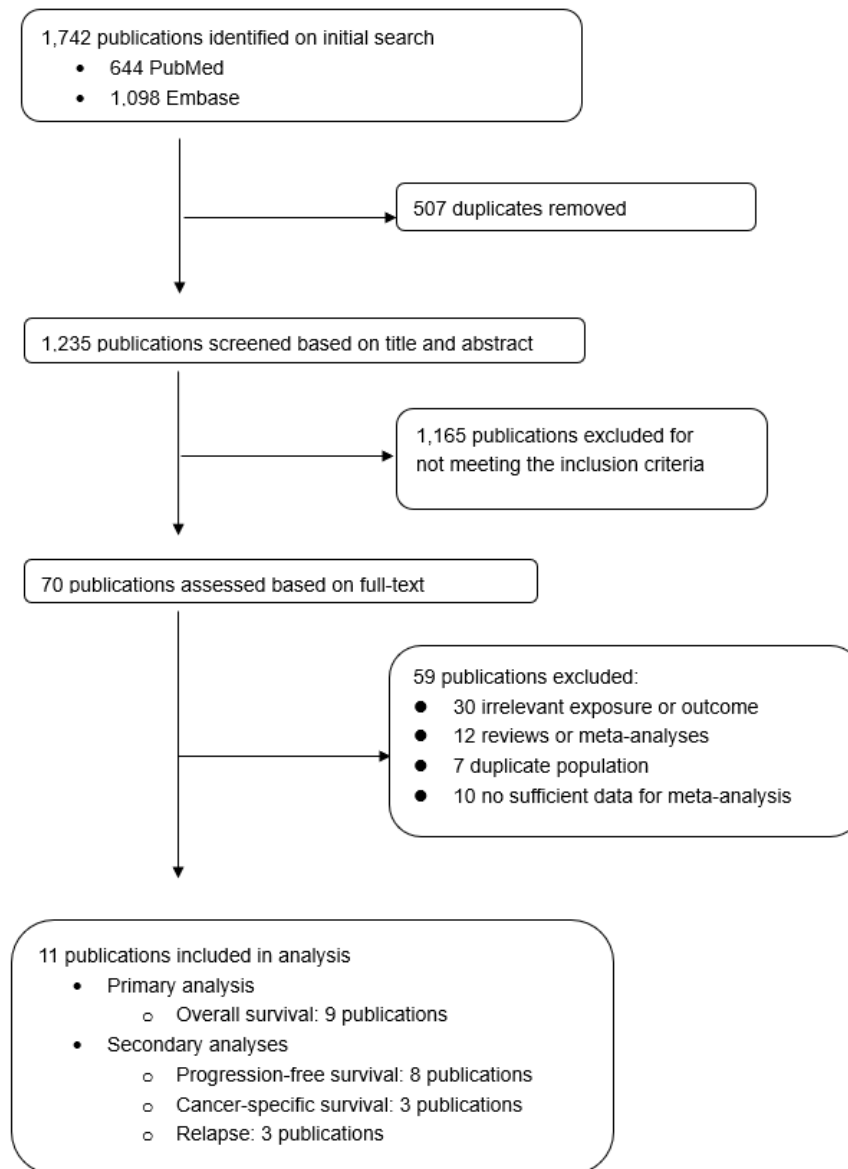
연구대상 및 방법

문헌 선택 과정

문헌 선택 및 제외 기준

✓ 선택	비타민 D 보충제 섭취 여부 (섭취 혹은 미섭취) 혹은 섭취하는 비타민 D 보충제의 비타민 D 성분 함량 차이 (고함량 혹은 저함량)를 기준으로 암 환자에서의 결과 지표를 비교한 연구
	OS, CS, PFS, Relapse의 결과 지표 중 하나 이상이 명시된 연구
	무작위 대조군 실험 (RCT, Randomized controlled trial) 연구 또는 코호트 연구(Cohort study)
✗ 제외	같은 코호트를 대상으로 진행 한 연구 (하나의 코호트를 대상으로는 하나의 연구만 선정)
	체계적 문헌고찰, 메타분석, 연구 계획서 혹은 동물 및 세포를 대상으로 하는 연구

문헌 선택 흐름도



연구대상 및 방법

데이터 추출, 통계 분석

데이터 추출

추출 대상

- 비타민 D 섭취군과 비타민 D를 섭취하지 않는 군의 정의 및 특성
- 핵심 질문에 해당하는 RR 값과 95% CI 값 및 환자의 수
- 투병 중인 암의 종류, 비타민 D 보충제 투약 시기, 환자 추적 기간, 연구 시작점에서 환자의 혈중 비타민 D 농도
- 이 외 연구의 저자와 시행 국가, 환자의 나이, 성별 등



통계 분석

분석 Tool: STATA 17 (StataCorp, College Station, TX)

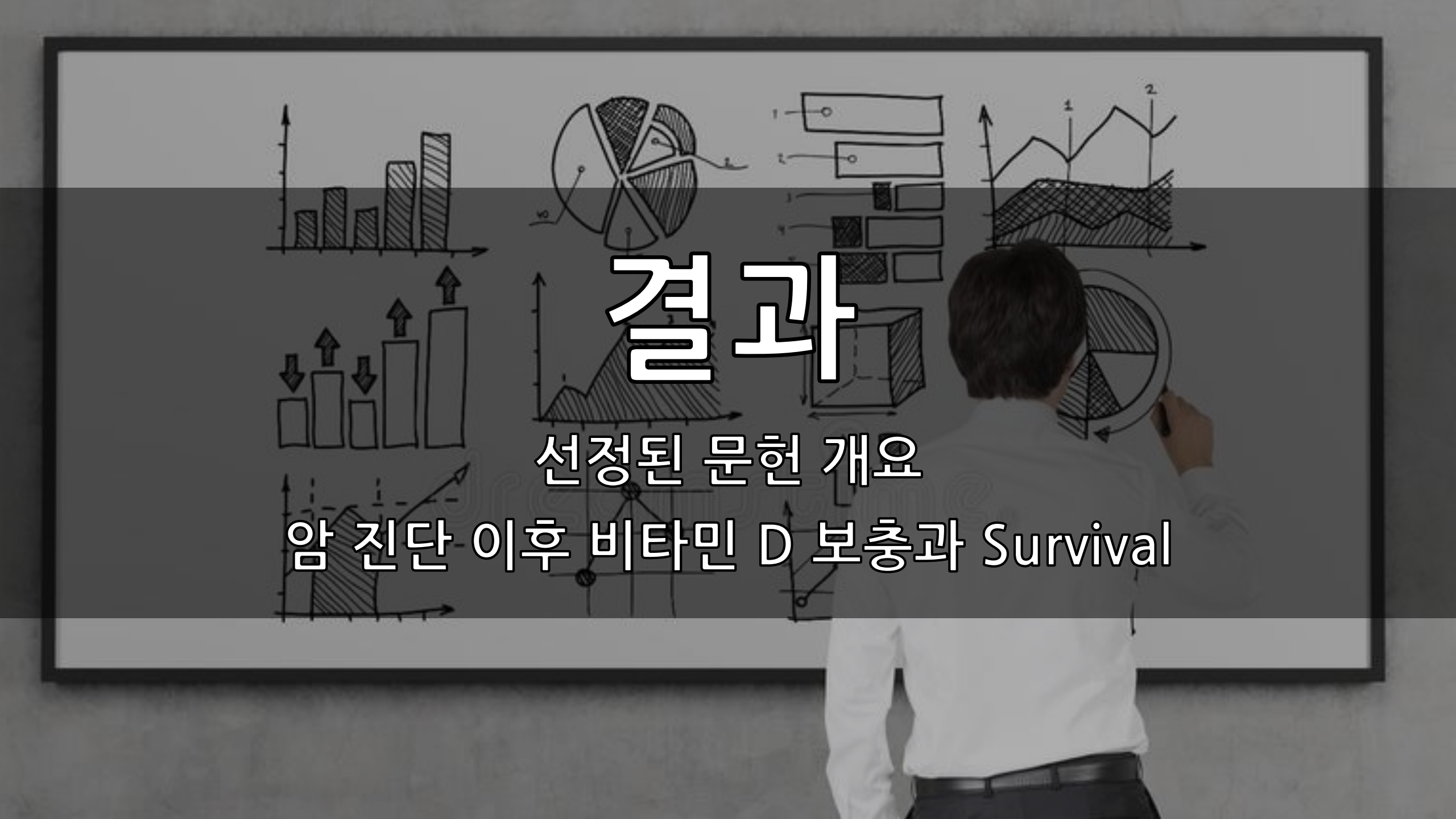
DerSimonian-Laird random effects model: OS, CS, PFS, Relapse RR 값, 95% CI값 계산

Egger's Test: 출판 비뮌임 확인 및 방지

I^2 : 각 연구간 이질성 척도 / 통계적 유의확률(**p-value**) 기준: 0.05

※ 단측검정 논문[Ng(2019)]: Avijit Hazra(2017)연구를 참고하여 양측검정(two-tail)으로 결과 산출





결과

선정된 문헌 개요

암 진단 이후 비타민 D 보충과 Survival

결과

선정된 문헌 개요

9 OS
(3+6)

8 PFS
(5+3)

3 CS
(1+2)

3 Relapse
(2+1)

5 RCTs

Supplementary Table 1. Main Characteristics of Studies Included				(3+6)	(5+3)	(1+2)	(2+1)	
Age			Timing of		Progression	Cancer-specific Surviv		
First author, Year, C	Trial name	Male (%)	Vit D supplementation	Overall Survival:	-free Survival:	al:	Relapse:	
ountry, Reference	Cancer patients	Follow-up period	Contrast for RR	RR (95% CI)	RR (95% CI)	RR (95% CI)	RR (95% CI)	Exclusion criteria regarding Vit D supplementation
Randomized Controlled Trials								
Urashima 2019 Japan (1)	AMATERASU	66	2-4 weeks after operation					
	digestive tract cancers (I-III)	66% median: 3.5 y	Vit D3 (2000 IU/d) vs. placebo during chemotherapy	0.81 (0.48, 1.36)	0.66 (0.43, 0.99)	1.09 (0.58, 2.01)	0.75 (0.48, 1.17)	Excluded current users taking Vit D supplement or active Vit D
Ng 2019 US (1)	SUNSHINE	56						
	advanced or metastatic colorectal cancer	57% median: 1.9 y	Vit D3 (8000/4000IU/d) vs. 400 IU/d	NA	0.64 (0.43, 0.96)	NA	NA	Excluded current users of taking 2000 IU/d or greater of Vit D3
Akiba 2018 Japan (1)	UMIN000001869	68	after operation (during chemotherapy)					
	Non-small cell lung cancer stage IA to IIIA	62% median: 3.3 y	Vit D3 (1200 IU/day) vs. placebo	1.22 (0.54, 2.79)	1.15 (0.64, 2.05)	NA	NA	Excluded current users taking Vit D supplement or active Vit D
Johansson 2021 Italy (1)	N.EudraCT 2009-012049-46	50.5	after operation (no adjuvant treatment)					
	recently resected, stage II melanoma	57% 1 y	Vit D3 (100,000 IU/50days) vs. placebo during treatment	NA	1.20 (0.37, 4.00)	NA	1.22 (0.32, 4.55)	Excluded current users of at least 600 IU/day of supplemental Vit D or high-dose calcium therapy within the prior 6 m
Golubic 2018 Croatia (1)	NA	69						
	metastatic colorectal cancer	51% total 46 m	Vit D3 (2000 IU/day) +standard treatment vs. standard treatment	1.01 (0.39, 2.61)	1.11 (0.69, 1.77)	NA	NA	Not specified

Abbreviations: M, male; m, month(s); NA, not available; Vit, Vitamin; y, year(s)

결과

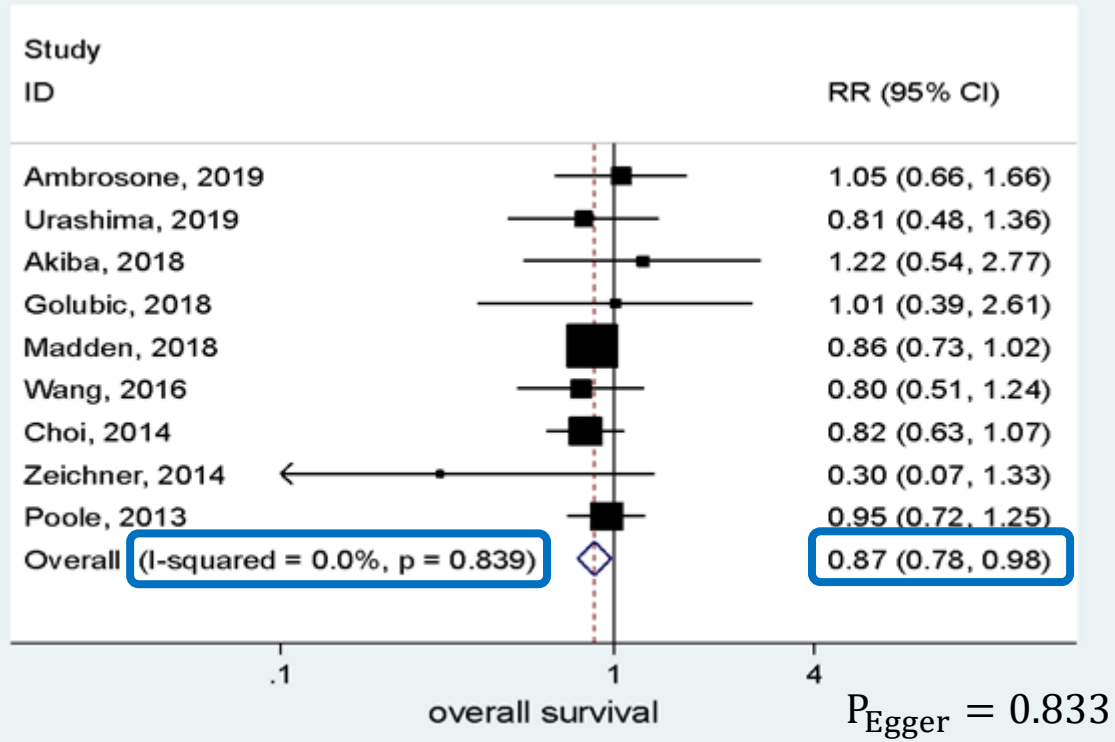
선정된 문헌 개요

6 Cohort Studies

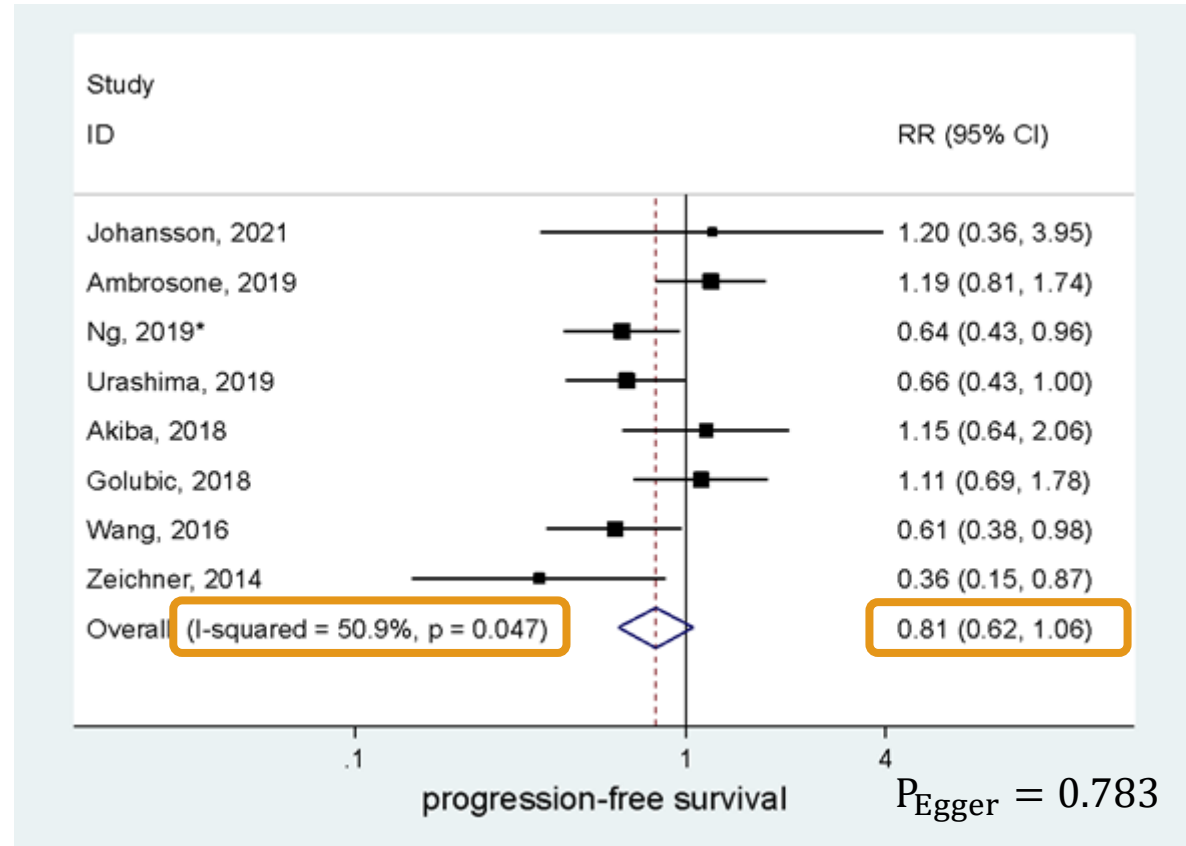
Cohort studies				9 OS (3+6)	8 PFS (5+3)	3 CS (1+2)	3 Relapse (2+1)	
Poole 2013 US+China (0)	After Breast Cancer Pooling Project	NA 0%	1-5 y after diagnosis	0.95 (0.72, 1.24)	NA	0.97 (0.68, 1.38)	0.92 (0.62, 1.35)	Not specified
	breast cancers (I-III)	NA	Users vs. non-users					
Ambrosone 2019 Mix (0)	DELCaP	0%	within 6 m (during chemotherapy)	1.05 (0.66, 1.65)	1.19 (0.81, 1.74)	NA	NA	Excluded users taking Vit D supplement before chemotherapy
	breast cancer	median 8.1 y	users vs. non-users					
Madden 2018 Ireland (0)	NCRI	NA	within 6m & after 6m after diagnosis	0.86 (0.72, 1.01)	NA	0.80 (0.64, .99)	NA	Excluded current users of taking Vit D supplement in the y prior to cancer diagnosis
	invasive breast cancer (I-III)	0%	users vs. non-users					
Zeichner 2014 US (0)	UM/SCCC	53 0%	during chemotherapy	0.30 (0.07, 1.37)	0.36 (0.15, 0.88)	NA	NA	Not specified
	HER2+ non metastatic breast cancer	median 29.5 m for D FS, 40.4 m for OS	users vs. non-users					
Wang 2016 China (0)	NA	NA 83%	after surgery during EC treatment and recovery phase	0.80 (0.51, 1.24)	0.61 (0.38, 0.98)	NA	NA	Not specified
	Esophageal cancer	2 y	users vs. non-users					
Choi 2014 US (0)	IWHS	78.9 0%	1-17 y after diagnosis	0.82 (0.63, 1.08)	NA	NA	NA	Not specified
	All cancers	mean 6.1y	Users vs. non-users					

결과

암 진단 이후 비타민 D 보충과 Survival



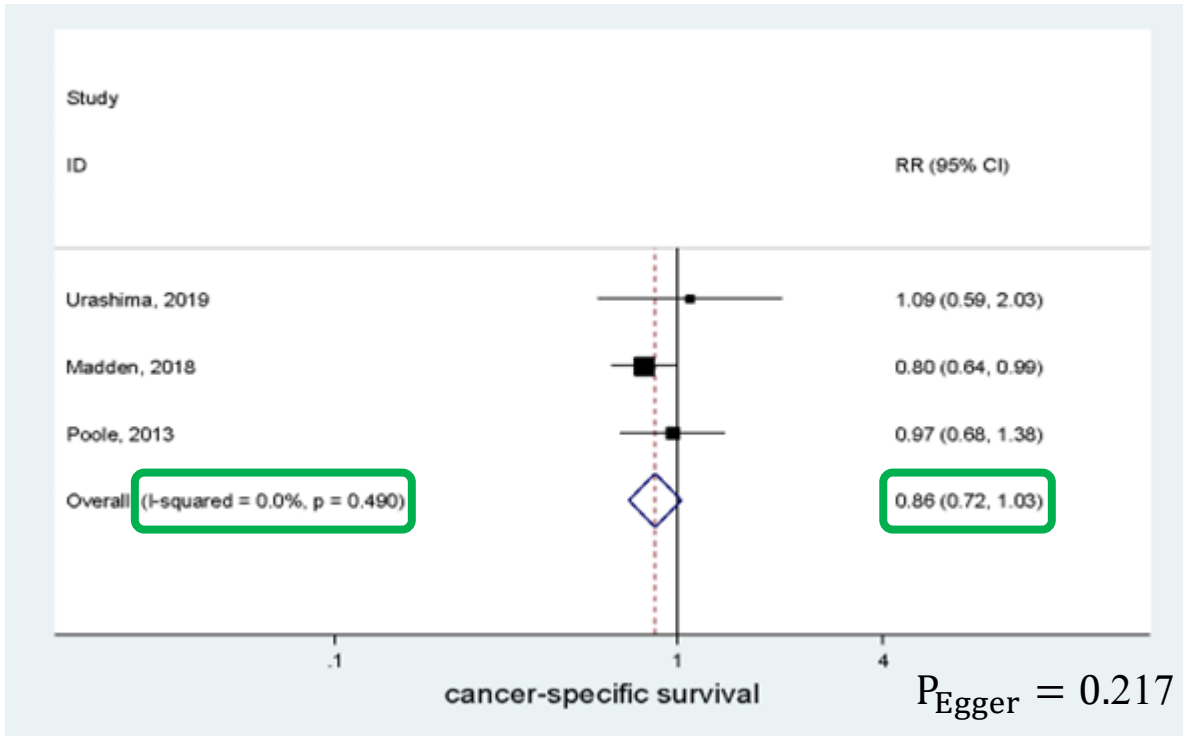
Overall survival



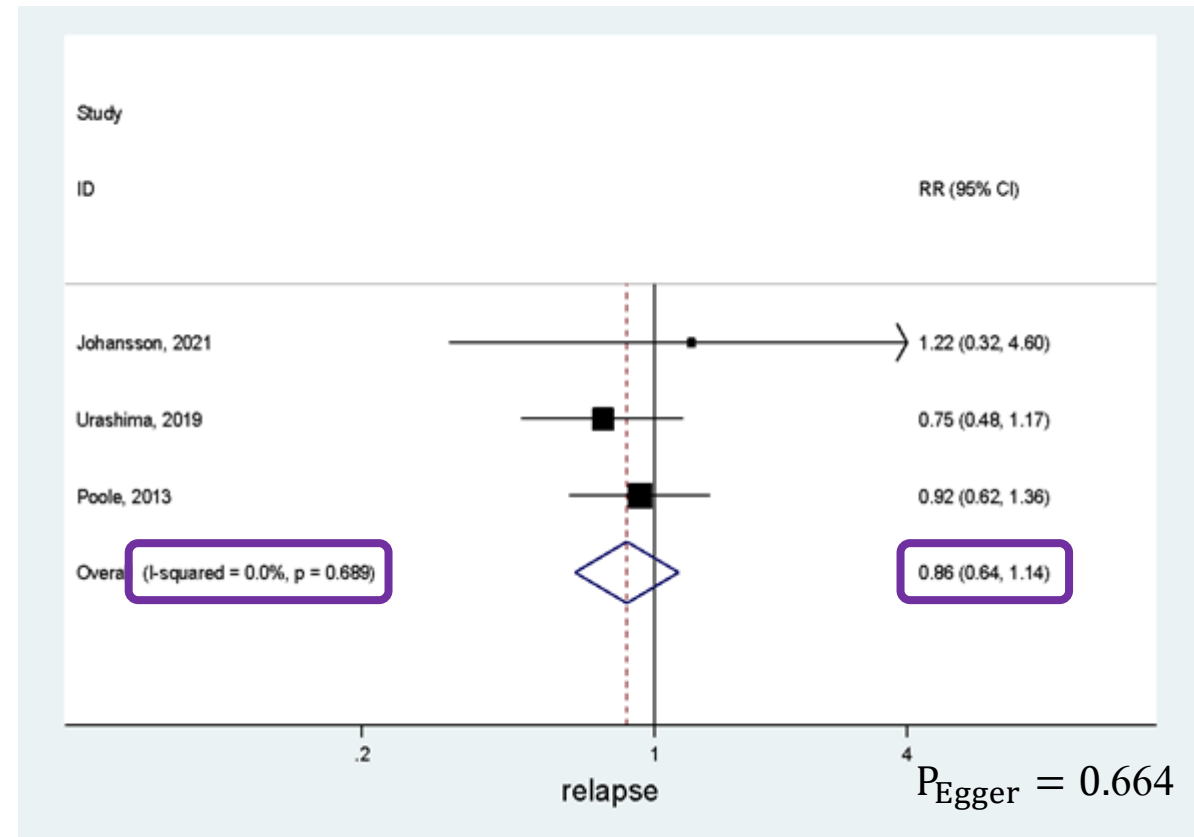
Progression-free survival

결과

암 진단 이후 비타민 D 보충과 Survival



Cancer survival

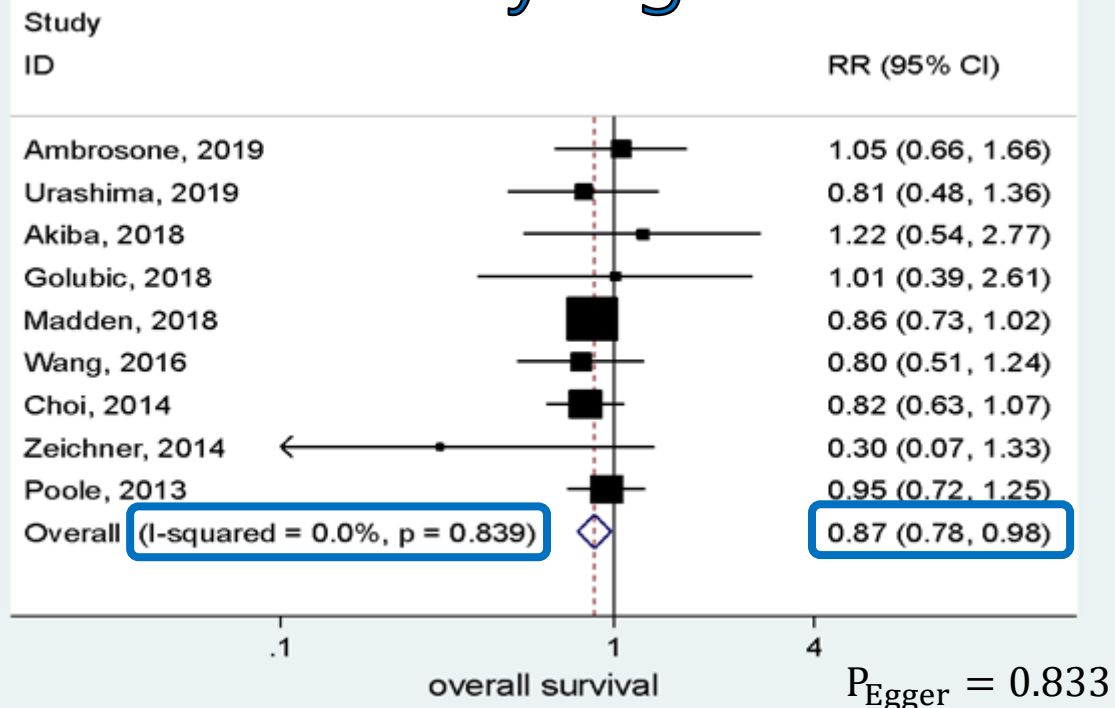


Relapse

고찰

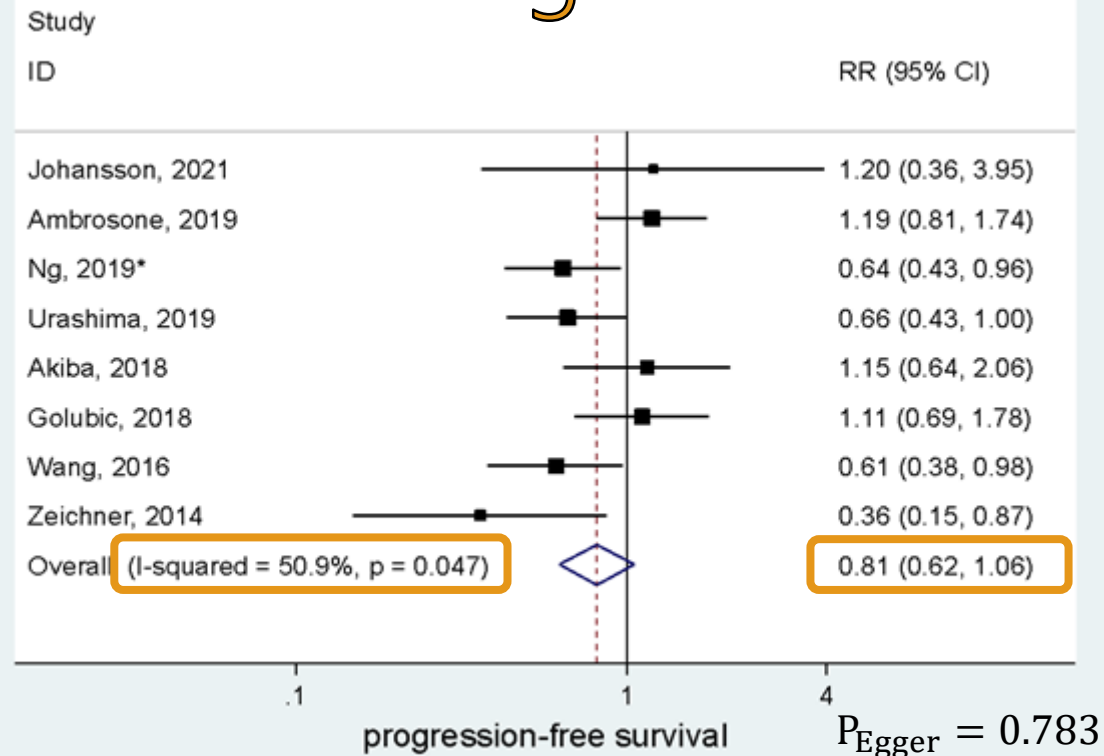
메타분석 결과 고찰

Statistically significant



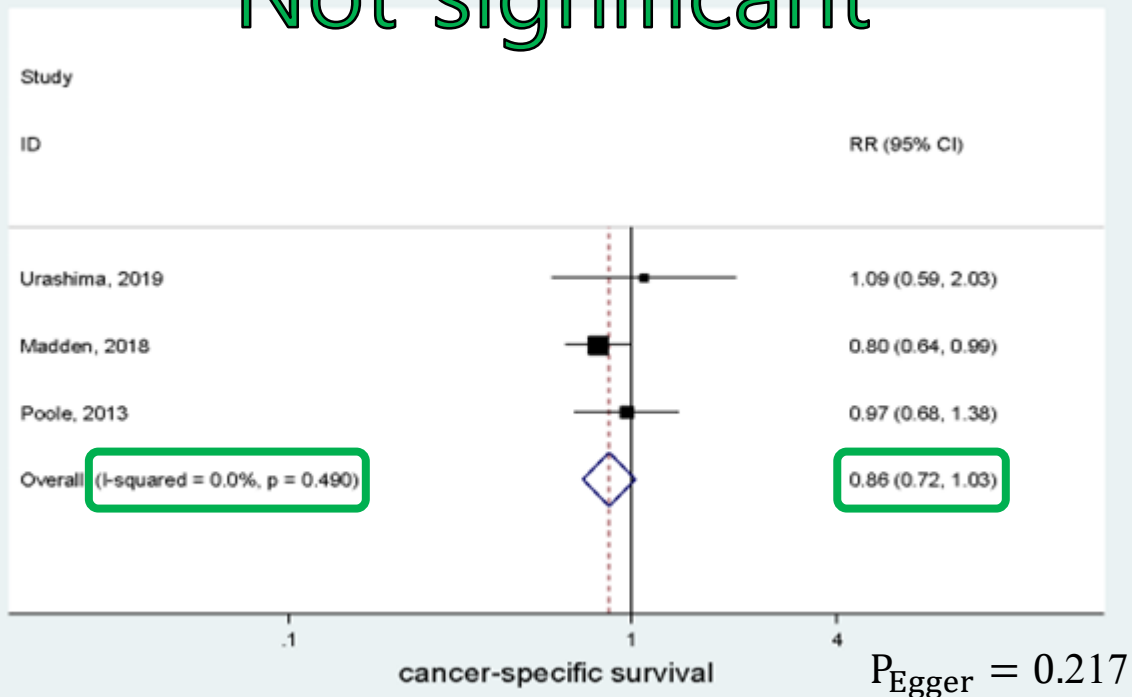
Overall survival

Not significant



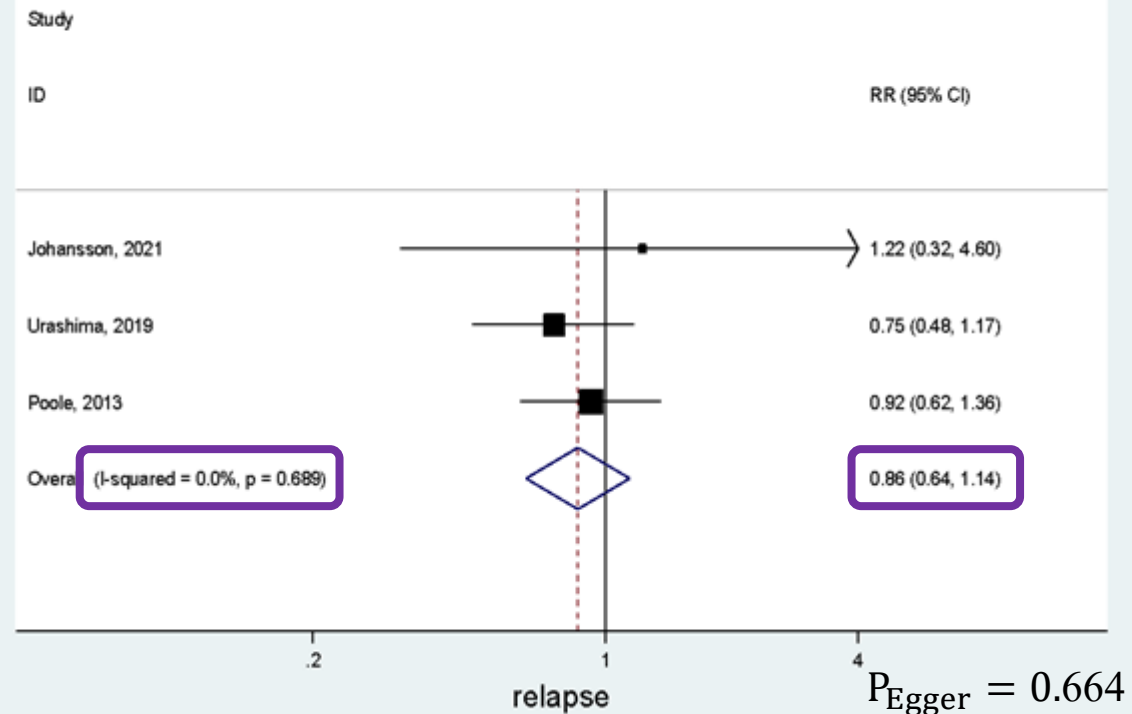
Progression-free survival

Not significant



Cancer survival

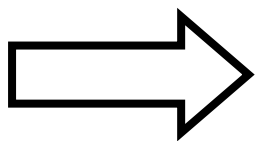
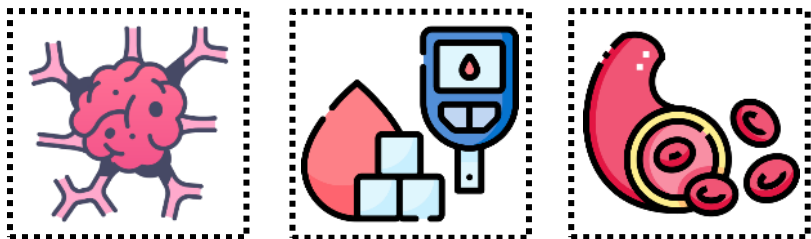
Not significant



Relapse

OS (Overall Survival)

환자가 치료를 받기 시작한 순간부터
사망한 순간까지의 기간



비타민D 보충제 섭취의
암 외 다른 질병 예방 가능성

Conclusions

Increases in cardiovascular disease and mortality have been associated with PHPT, although data in those with mild disease are lacking. Studies have attempted to understand the multifactorial mechanisms that might be responsible, such as hypertension, endothelial dysfunction, decreased vascular compliance, autonomic dysfunction, and left ventricular hypertrophy. Most

PHPT(Primary Hyperparathyroidism):
부갑상선 기능 항진증

그림 1

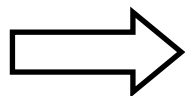
with elevated or high normal PTH levels. In our opinion, in particular, vitamin D supplementation is a promising therapeutic approach. Apart from lowering PTH levels, vitamin D is suggested to exert numerous beneficial effects on the cardiovascular system, prevents fractures and falls, and may reduce other diseases such as infections and cancer.^{17,30-33} Importantly, treatment of secondary

그림 2

부갑상선 호르몬: 혈압 증가, 심장비대, 부정맥 등 **심장 질환** 유발



비타민 D: 부갑상선 호르몬의 작용 **억제**



비타민 D: 심장 질환의 위험 예방 가능

brane Ca^{2+} channels in mediating rapid effects of the hormone. We propose that the $1,25\text{-(OH)}_2\text{D}_3$ -mediated Ca^{2+} signaling pathway may be involved in the regulation of insulin secretion from the pancreatic β -cell. (*Endocrinology* 136: 2852-2861, 1995)

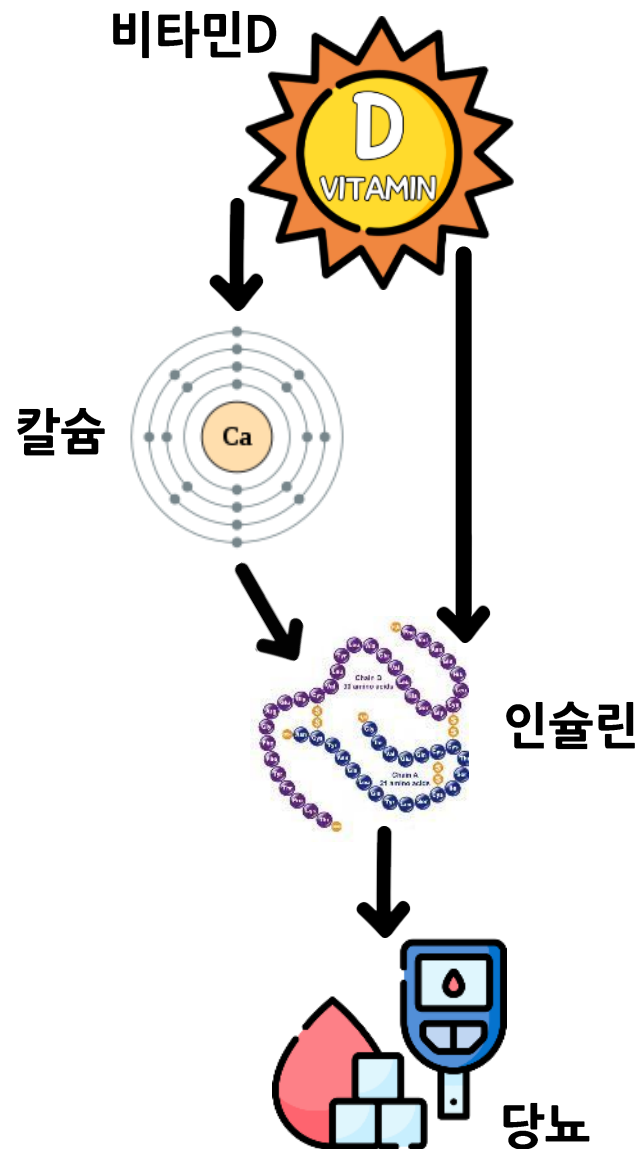
그림3

mal. The calculated insulin area during an oral glucose load was significantly correlated with serum calcium (5.1 to 12.2 mg per deciliter), and a linear relation was obtained ($y = 1.59x - 3.3$, $r = 0.81$, $p < 0.001$), although a relation with the glucose area was not found. These observations indicate that serum calcium has an important effect on insulin secretion in parathyroid disorders. (*N Engl J Med* 292:501-504,

그림4

tion. Optimal secretion of insulin occurred at an extracellular calcium concentration of 2.64 mM. The omission of calcium inhibited glucose-stimulated insulin secretion from pancreas of 27 day rabbit fetuses.

그림5



선행 연구 1

Nevertheless, it is uncertain whether supplementation with vitamin D prevents cancer or cardiovascular disease, because such results cannot establish causality.^{1,4,7,8} For example, observational

그림8



The association between vitamin D and cancer has long been studied, but the results have been variable. Thus, there does not seem to be a consensus on whether vitamin D has a beneficial anti-cancer effect. This review not only summarizes the association between vitamin D and cancer risk

그림6 그림9

13% reduced cancer mortality over 3-10 years of follow-up, which confirms findings in the previous meta-analysis in 2014 [24]. Corresponding to the 13% reduction in cancer mortality,

추가 연구 필요

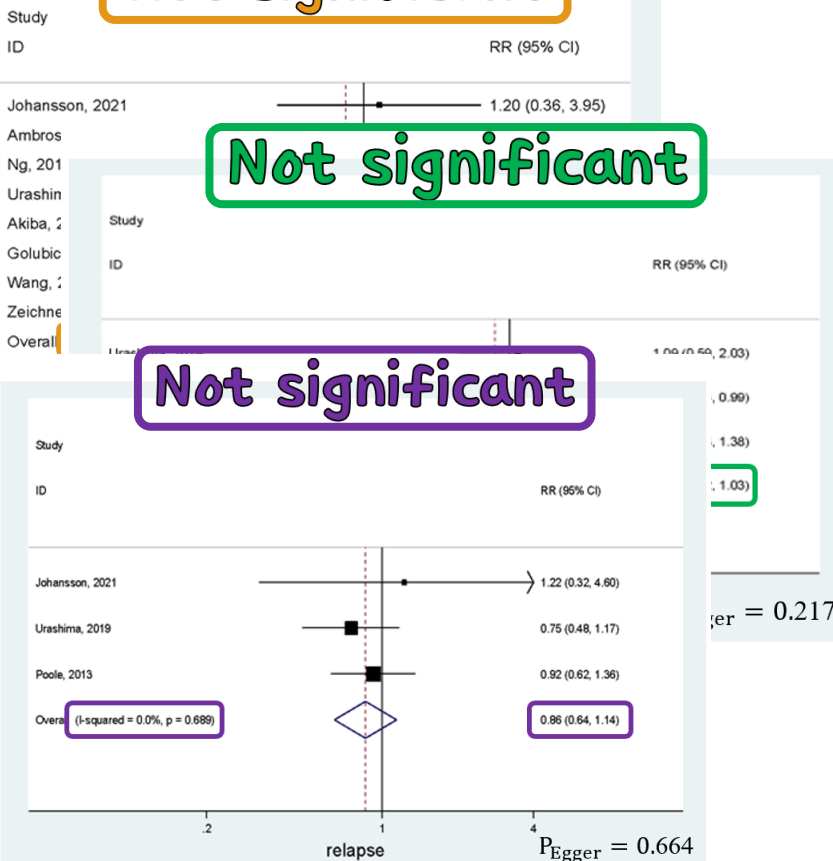
비타민 D의 암에 대한 생리... 그림7 등

본 연구

Not significant

Not significant

Not significant





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참고문헌

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감사합니다.

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